

Group: A
Meeting Number: 2
Date and Time: 19/01/2026: 19:00

Present:

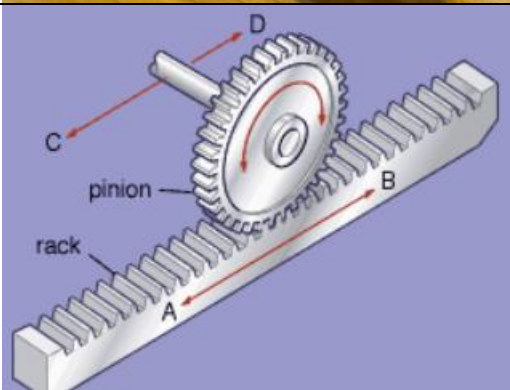
Balraj Bhuller
 Christopher Mead
 Hannah Arnold
 Max McGill
 Rachel Kirkham

Items for discussion:

Prepare for validation meeting on the first week back of university, 27/01/2026.

Discuss a rough basic idea on the Soapamorph prototype design, constraints and functionality.

Record of discussion

Foam sheets – mould head. • 3d Scan for 3d printing or use the haptic pen.	
Peristaltic Pump • Blender design for the mouth tongue track. • One way pump motion.	
Track motion for inner head • Gear track – for movement Or linear actuator	
Silicone	Coating or base to keep bacteria out.

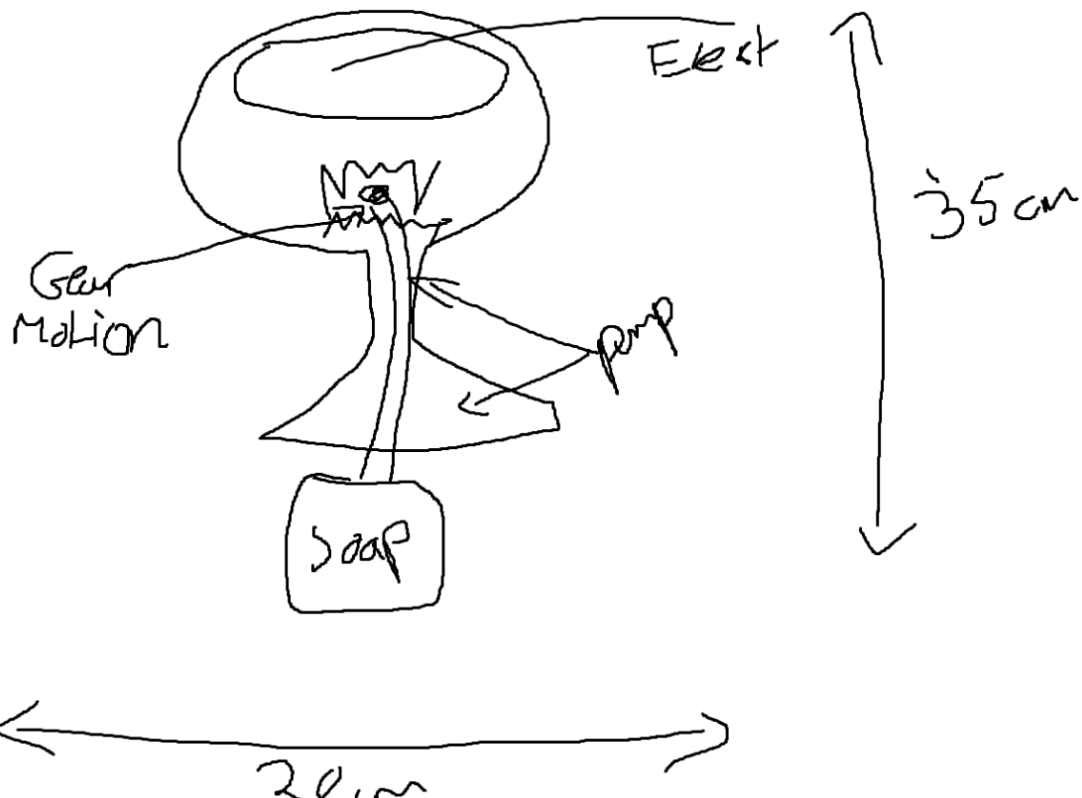
Non – porous surface meaning little opportunity for bacteria to anchor.
Doesn't cancel the risk fully of bacteria lingering however its more efficient at keeping bacteria at bay.
Easier to clean and maintain.

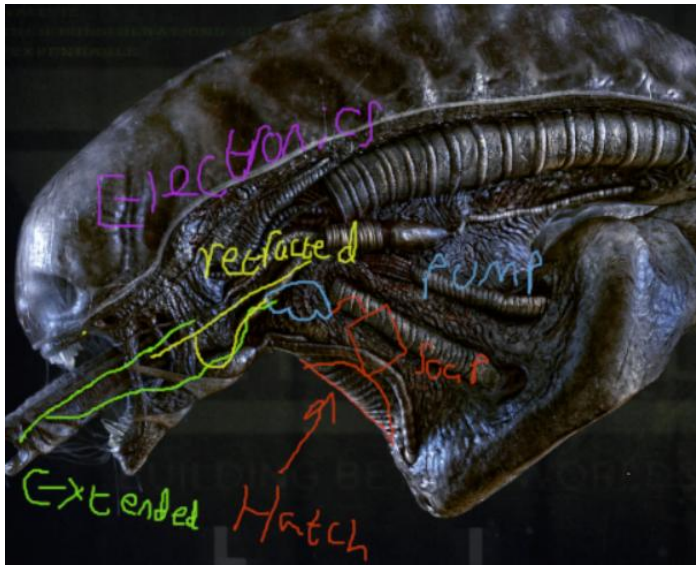
Processes to show:

- Blender design.
- 3d printing.
- 3D Scanning or Pen Haptics.
- Possible silicone processing for anti-bacterial design.
- Last but not essential, electronic design and implementation.

Key things to consider:

- Possible skeleton design if silicone base.
- Non-hollow tongue for tube alignment.
- Attachable tongue for cleaning.
- Electronics not coming into contact with moisture or soap.
- Electronics pushed back to the end.
- Weighted base – Metal attachment around the base to secure weight
- Electronics not being lower than the soap just in case of leakages.





Next Steps:

Action No.	Action Required	Success criteria	By When	By who
1.	Individual research: What motion techniques to use for the tongue mechanism.	100%	27/01/2026	ALL
2.	Individual research: What pump method and materials would be best other than what we discussed.	100%	27/01/2026	ALL
3.	Individual research: Material research for entire model base, justified why it would work better than what was discussed or why what is discussed works best.	100%	27/01/2026	ALL
4.				
5.				
6.				